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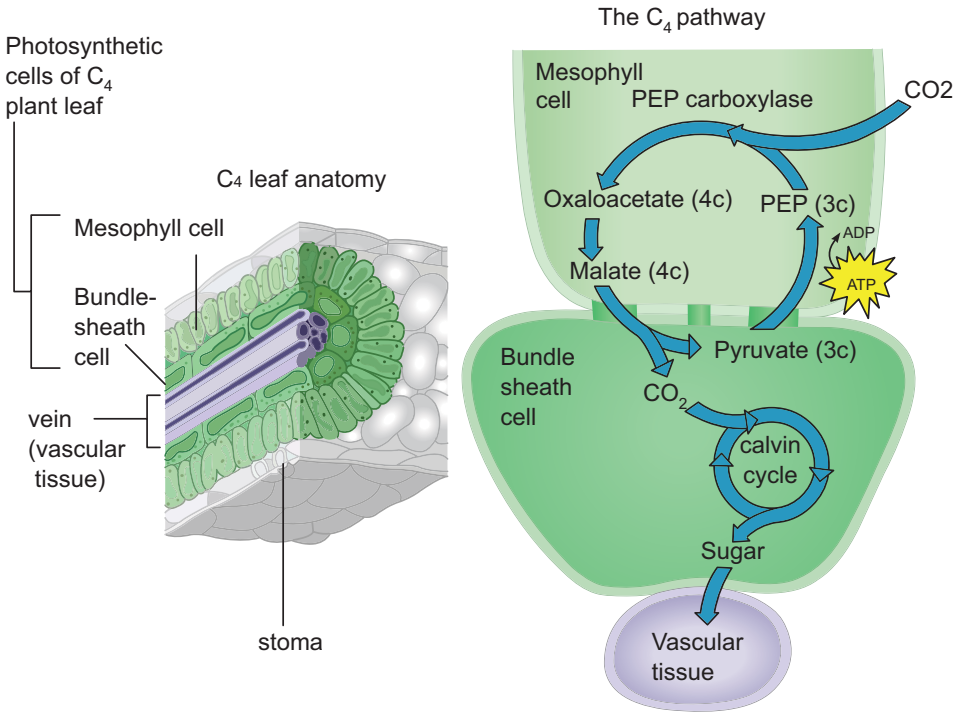


Figure 13.20: C_4 Cycle

Earth's plant biomass and 1% of its known plant species. Despite this scarcity, they account for about 30% of terrestrial carbon fixation. Increasing the proportion of C_4 plants on earth could assist biosequestration of CO_2 and represent an important climate change avoidance strategy.

C_4 pathway is completed in two phases, first phase takes place in stroma of mesophyll cells, where the CO_2 acceptor molecule is 3-Carbon compound, phospho enol pyruvate (PEP) to form 4-carbon Oxalo acetic acid (OAA). The first product is a 4-carbon and so it is named as C_4 cycle. oxalo acetic acid is a dicarboxylic acid and hence this cycle is also known as **dicarboxylic acid pathway** (Figure 13.20). Carbon dioxide fixation takes place in two places one in mesophyll and another in bundle sheath cell (di carboxylation pathway). It is the adaptation of tropical and sub tropical plants growing in warm and dry conditions. Fixation of CO_2 with minimal loss is due to absence of photorespiration. C_4 plants require 5 ATP and 2 NADPH + H^+ to fix one molecule of CO_2.

13.14.1 Stage: I Mesophyll Cells

Phosphoenol Pyruvate + CO_2
(PEP) (3C)
PEP carboxylase
↓
Oxaloacetic acid (OAA) (4C)

Oxaloacetic acid (OAA) is converted into malic acid or aspartic acid and is transported to the bundle sheath cells through plasmodesmata.

13.14.2 Stage: II Bundle Sheath Cells

Malic acid undergoes decarboxylation and produces a 3 carbon compound Pyruvic acid and CO_2. The released CO_2 combines with RUBP and follows the calvin cycle and finally sugar is released to the phloem. Pyruvic acid is transported to the mesophyll cells.